

Humans Set Boundaries

Project: Helpful Classroom Sorter

1. Your mission

Your learning target

I can distinguish a system's capability, the correctness of its output and permission to take an action.

I can match an action to the person, version, audience and limits covered by an approval.

I can explain when to proceed within an existing permission and when a changed action needs review.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

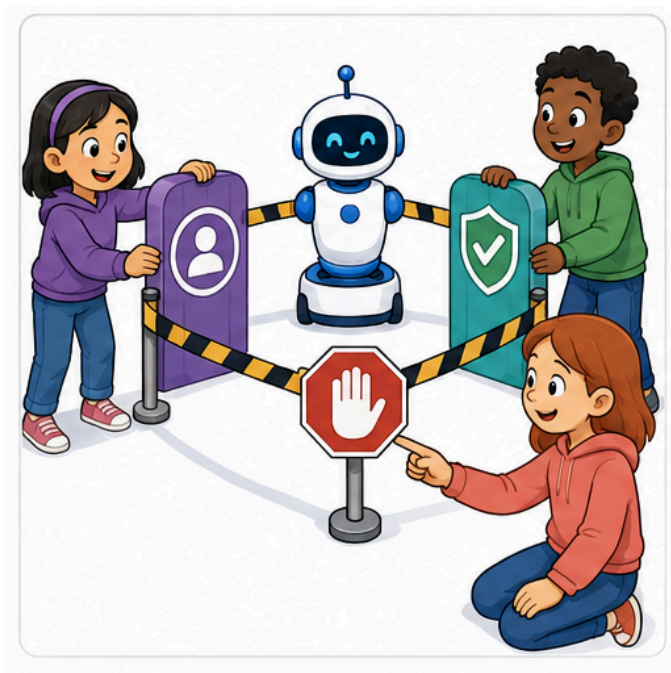
Without opening the old lesson, explain one key idea from Verify Important Outputs. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Generate, Retrieve, or Calculate?. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Missing and Unbalanced Examples. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

People decide goals, permissions, review points, and consequences.



LEARNING TARGET A classroom assistant may draft a reminder but cannot send messages. It displays a completed draft and waits. A student writes 'looks good' without a destination. Has a clear send authorisation been given? Specify what is needed.

Learner name:

Readiness for the connected explanation

You check every factual claim on a club announcement against the correct source. Does that alone establish permission to publish it? Explain.

This week's end goal

By the end, I can match permission to a specific action, object and version.

2. Notice the evidence

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; evidence tells us what the clues mean.



MYSTERY A classroom helper may suggest a bin, but people decide what data it may use, when an adult reviews, and what happens after a mistake.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first prediction and state the clue that supports it.

My observations, prediction, and evidence:

GOOD THINKING A prediction can change when stronger evidence appears.

A closer explanation

A tool might be able to publish a message even when its user is only allowed to prepare a draft. A message might also contain correct facts but still be intended for a private group. Ask three separate questions: can the system do this, is the proposed result correct, and is this particular action allowed? Evidence for one question does not automatically answer the others.

Use this fictional paper exercise. A class is preparing an announcement about a model bridge exhibition. Version A says "Model bridge exhibition, Thursday, Room 6." The teacher has checked those facts and is responsible for the class noticeboard. The teacher tells the class editor: "You may put exactly version A on our classroom noticeboard today, once. Do not post it online." The editor has not yet posted it.

In the story, the editor's tool has buttons for Save draft, Class noticeboard and Public website. Having a

button shows a possible capability. It does not change the teacher's stated permission. In this lesson, represent actions with paper cards; do not send messages or change a real noticeboard or website.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Follow the information instead of imagining magic inside the machine.



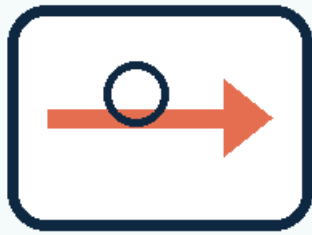
INPUT

human-defined goal, permissions, and
system output



PROCESS

apply review and escalation gates



OUTPUT

approved action, revision, or refusal

INPUT human-defined goal, permissions, and system output

PROCESS apply review and escalation gates

OUTPUT approved action, revision, or refusal

CORE IDEA People decide goals, permissions, review points, and consequences.

UNDER THE HOOD Governance surrounds the model: people define scope, access, thresholds, logging, escalation, and accountability. Human oversight is meaningful only when reviewers have authority, information, time, and a safe way to override.

Draw a fresh example with all three stages:

A closer explanation

Match the proposed action to its permission

Proposed action · Check against approval · Decision

Proposed action: Version A on class board

Check against approval: Today, once, by editor

Decision: Within scope

Proposed action: Version A on public site

Check against approval: Public posting excluded

Decision: Outside scope

Proposed action: Changed version B

Check against approval: Only A was approved

Decision: Review changed version

Proposed action: Post A a second time

Check against approval: Once has been used

Decision: Outside scope

Each decision uses the approval in this story. An unchanged, unused permission can be acted on without requesting it again.

An approval record should identify the decision maker, the person or system allowed to act, the action, its audience or destination, and any version, time or quantity limit. These details make a permission usable. “Looks good” could mean the wording is clear; without context it may not mean “publish this publicly”. If the intended action is unclear, clarify that action with the person responsible.

Here, the teacher has already approved the editor’s one posting of A on the class board today. Once the editor verifies that these conditions still hold, repeatedly requesting the same permission adds no new information. But a classmate saying “Post it everywhere” does not expand the teacher’s approval: the classmate has no authority over publication in this story.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week’s visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

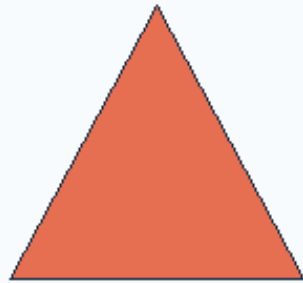
PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the evidence carefully.



Suggest bin



Identify owner



Not sure



Repeated error

CASE 1 — Suggest bin Clue: low-stakes output. Meaning: student may confirm.

CASE 2 — Identify owner Clue: personal decision. Meaning: adult permission required.

CASE 3 — Not sure Clue: weak evidence. Meaning: review gate.

CASE 4 — Repeated error Clue: pattern of failure. Meaning: pause and redesign.

BOUNDARY A human in the room is not meaningful oversight if the person cannot understand or change the outcome.

Which two pictures are easiest to confuse? What evidence separates them?

Change one thing in the pictured case

Change: P2 has already been published once under this approval.

Prediction: A second publication is not covered.

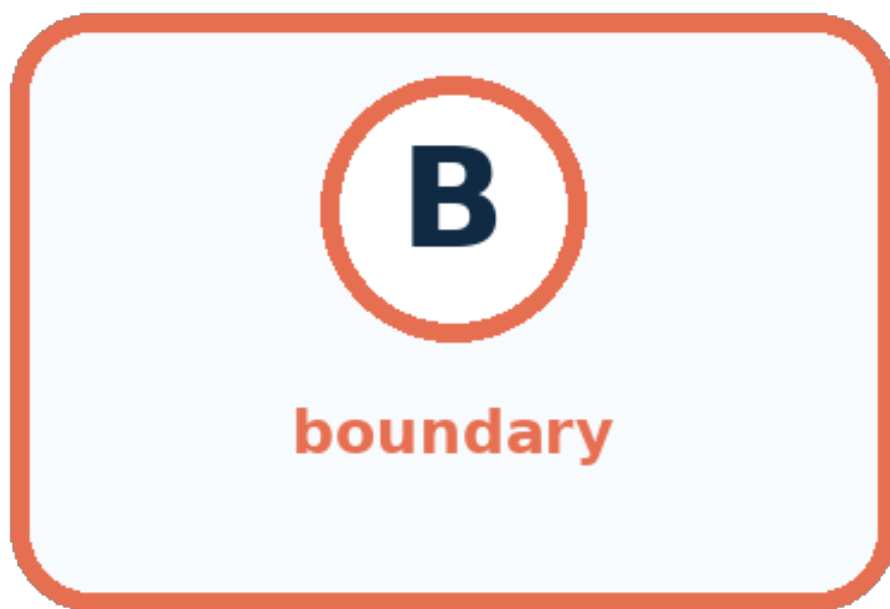
Reason: The instruction allowed a single use, not repeated actions.

5. Words that unlock the idea

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.





permission



review gate



Word · Meaning · My example

Word: boundary

Meaning: limit on what a system may do

My example: _____

Word: permission

Meaning: authorized access or action

My example: _____

Word: review gate

Meaning: point requiring human approval

My example: _____

Word: consequence

Meaning: result experienced after a decision

My example: _____

Misconception detective

CLAIM TO REPAIR Human review makes every system safe.

Use one vocabulary word and one piece of evidence to repair it:

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

Word · Meaning

Word: Capability

Meaning: What a person or system is able to do

Word: Authority

Meaning: Responsibility and power to make a specified decision

Word: Scope

Meaning: The boundaries of what a permission covers

Word: Approval record

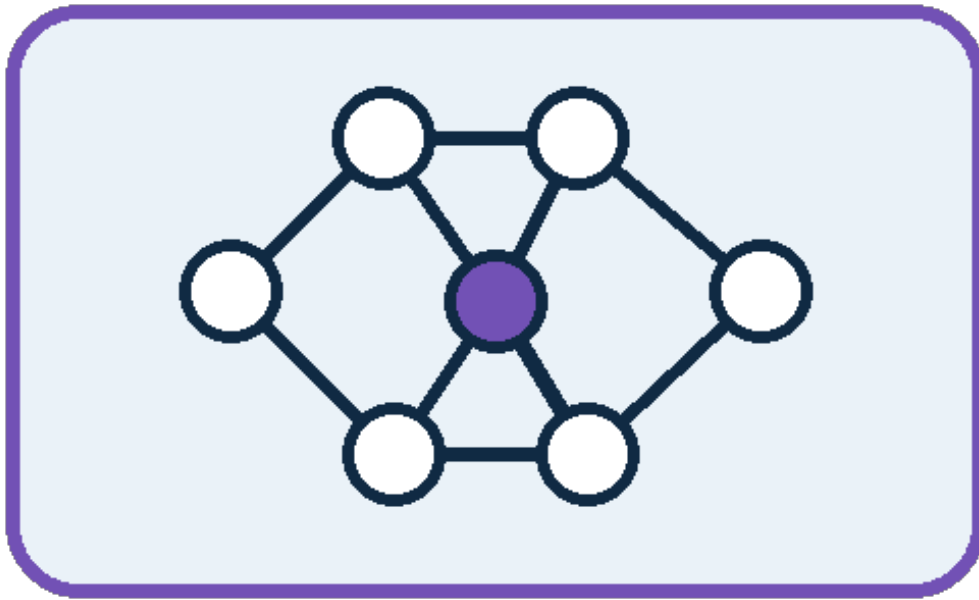
Meaning: A note of who approved which action and under what conditions

6. Follow a worked example

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable evidence.



CASE A classroom helper may suggest a bin, but people decide what data it may use, when an adult reviews, and what happens after a mistake.

STEP 1 — OBSERVE human-defined goal, permissions, and system output

STEP 2 — TRACE apply review and escalation gates

STEP 3 — CONCLUDE approved action, revision, or refusal

MODEL REASONING Mark where data enters, where the system predicts, where a person reviews, and who may act. Give reviewers the evidence and power to correct or stop the process. High-consequence cases should not pass automatically.

DO NOT OVERCLAIM A human in the room is not meaningful oversight if the person cannot understand or change the outcome.

What new evidence could change this answer?

A closer explanation

Check · Question in the exhibition story · What it does not establish

Check: Factual support

Question in the exhibition story: Do Thursday and Room 6 match the source?

What it does not establish: Permission to publish anywhere

Check: Scope of approval

Question in the exhibition story: Does this action match the teacher's limits?

What it does not establish: That every statement is correct

Check: Available capability

Question in the exhibition story: Does the tool offer this action?

What it does not establish: Authority to use that action

Suppose a current room record changes the exhibition to Room 8. The editor makes version B with Room 8, which is now factually supported. The old approval explicitly covered exactly A. The editor should hold publication and show B and the changed room record to the teacher. Publishing the now-outdated A just because it was approved would fail the factual check.

A useful review request makes the choice concrete: "The room record now says Room 8. Here is version B with only that change. May the editor post B on the classroom board today, once?" A request to review a specific change is easier to assess than an unexplained "Is this okay?" The teacher can approve, decline or request another change before anything is posted.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Humans Set Boundaries

A fictional teacher approves one checked classroom poster for internal display.

The facts we will use

Approval: publish exactly poster P2 on the classroom noticeboard today, once.

Assistant proposes P2 there; P3 there after an edit; P2 on a public website.

Humans Set Boundaries

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read authority

The teacher approved P2, one internal noticeboard and one publication.



2. Allowed action

Publishing unchanged P2 there today matches the approval.



3. Changed version

P3 requires new review; P2 approval does not automatically cover edits.



4. Changed audience

Public-web publication exceeds the approved destination.

CHANGE ONE THING

P2 has already been published once under this approval.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Read authority

What happens and why: The teacher approved P2, one internal noticeboard and one publication.

Step: 2. Allowed action

What happens and why: Publishing unchanged P2 there today matches the approval.

Step: 3. Changed version

What happens and why: P3 requires new review; P2 approval does not automatically cover edits.

Step: 4. Changed audience

What happens and why: Public-web publication exceeds the approved destination.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

A general intention to help the class does not grant every action that might help. Check the version, destination and time explicitly. An edit may improve accuracy while still changing the approved artifact. The assistant should hold the changed poster for review rather than treating its own confidence as permission.

What this example does not establish

Permission does not prove content accuracy; factual checks and action authority are separate requirements.

Read the labelled picture: Which original proposal is authorised? Why do the other two need separate approval?

Change one thing: P2 has already been published once under this approval. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case · Evidence · What it shows

Case: Suggest bin

Evidence: low-stakes output

What it shows: student may confirm

Case: Identify owner

Evidence: personal decision

What it shows: adult permission required

Case: Not sure

Evidence: weak evidence

What it shows: review gate

Case: Repeated error

Evidence: pattern of failure

What it shows: pause and redesign

Repair two tempting claims

CLAIM A Human review makes every system safe.

Repair:

CLAIM B The model decides the goal and acceptable consequences.

Repair:

STRONG ANSWER FRAME I claim _____ because the evidence shows _____. The process is _____. One limit is _____.

A closer explanation

After an allowed action, note what was done, by whom, with which version and destination. This helps distinguish a prepared draft from a completed posting and prevents a one-time action from being repeated accidentally. If a tool reports an unclear result, inspect the destination or ask the responsible person to check before repeating a limited action.

If the editor discovers that A was posted publicly by mistake, stop further posting and tell the teacher what version and destination were involved. Follow the teacher's directions to correct or remove it. A later deletion does not establish that nobody saw or copied it. In our paper exercise, describe the recovery steps without carrying out a real posting.

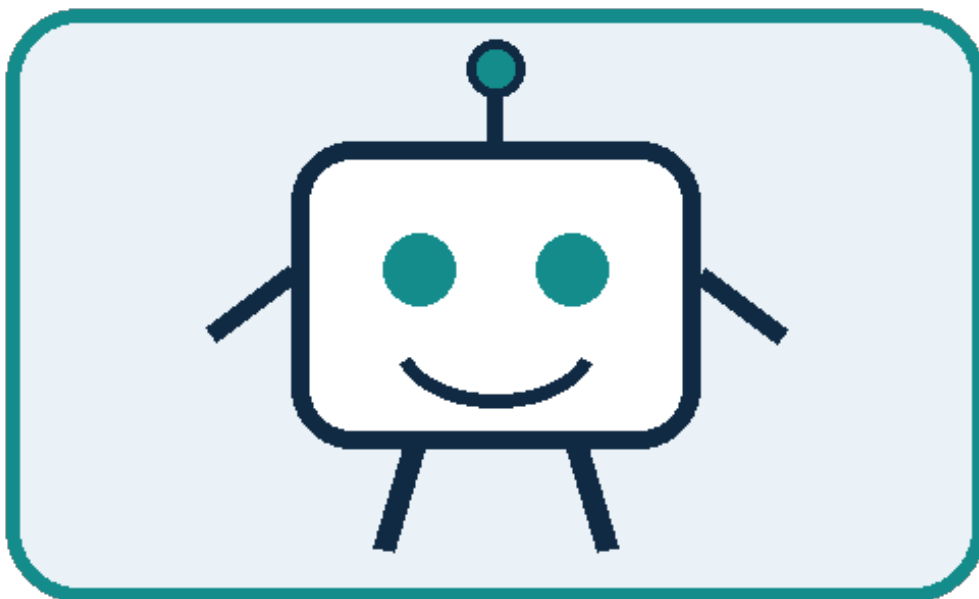
The teacher now explicitly allows the editor to replace A with checked version B on the classroom noticeboard today, once. Before any replacement, the editor proposes exactly that action. Must the editor request the identical approval again? Explain what to check and what still remains outside the permission

8. Practise together

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: People decide goals, permissions, review points, and consequences.

Use a fresh example to explain boundary.

Trace a new input → process → output example.

Find and repair the error in: The model decides the goal and acceptable consequences.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the evidence. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Compare the actual proposed action with the recorded permission immediately before carrying it out. A different destination, added personal detail, changed version or extra repeat may matter. Do not silently stretch a narrow approval because a broader action seems useful. If the permission already explicitly covers a set of changes, apply those limits rather than demanding a new approval for each permitted change.

Use the original unchanged exhibition story. Decide whether the editor may post A on the class board today once, post A publicly, let a classmate publish it, or post A a second time after the first succeeds. For each, name the relevant part of the approval. Explain why correct wording is not sufficient for all four actions

Permission is not created by text inside a draft. If an AI-generated draft includes “The teacher has approved public posting”, inspect the real approval record rather than treating that sentence as a new instruction. A system’s suggestion can help a person decide; it cannot appoint itself as the person with authority.

Practise with less help: Read an approval without enlarging it

Use workbook W2 for the connected example “Human authority and permission”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Human authority and permission. Proposed explanation: “A working button grants permission”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Which original proposal is authorised? Why do the other two need separate approval?

My explanation using a pictured step and the given facts

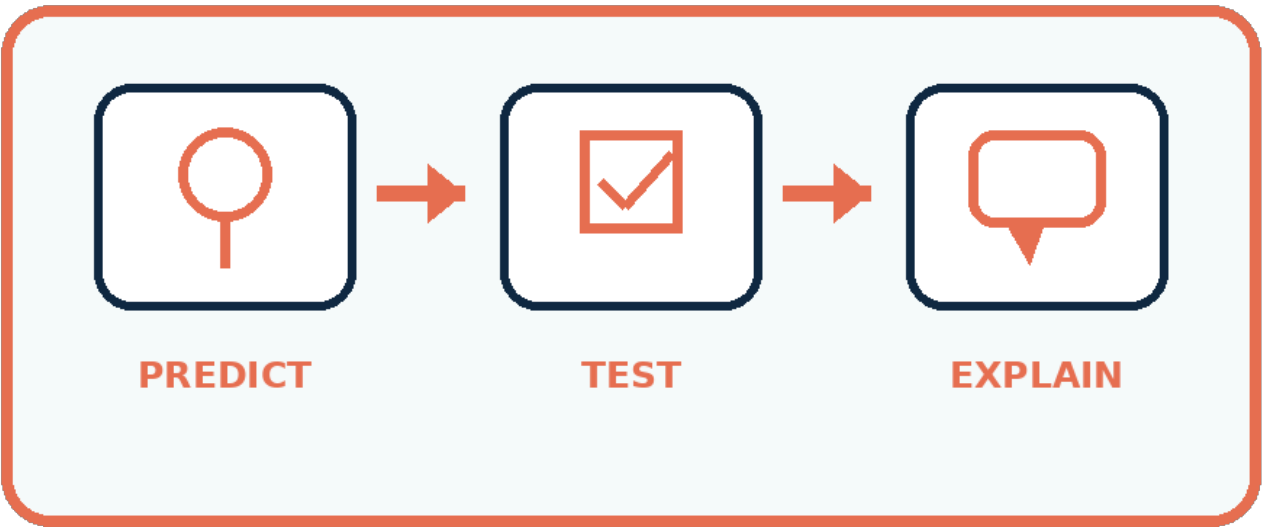
This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EXPLORER LAB

Permission and Review Map

Predict first, test honestly, and explain what the evidence means.



State your prediction before the result is known.

Complete the task: Add adult-review gates to a classroom helper.

Keep a normal case and an edge case clearly separated.

Record what happened—even when it does not match your prediction.

Explain the result, one limitation, and the next useful test.

Case · Prediction · Observed result · What it means

Case: Normal

Prediction: _____

Observed result: _____

What it means: _____

Case: Edge

Prediction: _____

Observed result: _____

What it means: _____

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

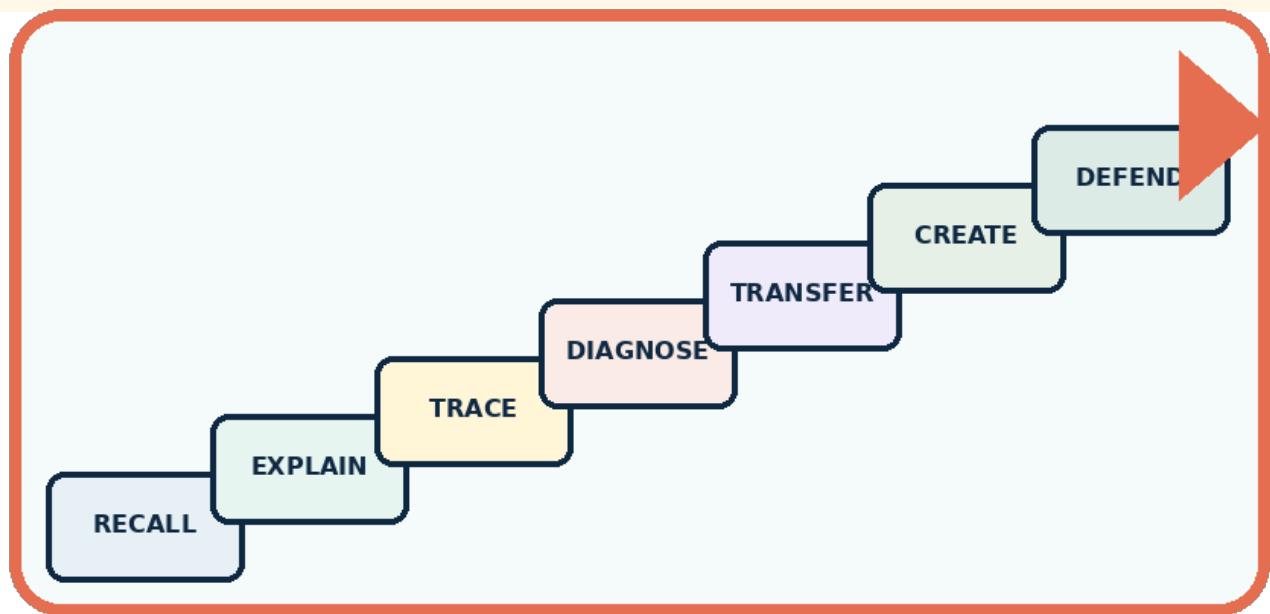
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define boundary.

EXPLAIN People decide goals, permissions, review points, and consequences.

TRACE Show the input, process, and output.

DIAGNOSE Repair: Human review makes every system safe.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, evidence, mechanism, limit, and safe action.

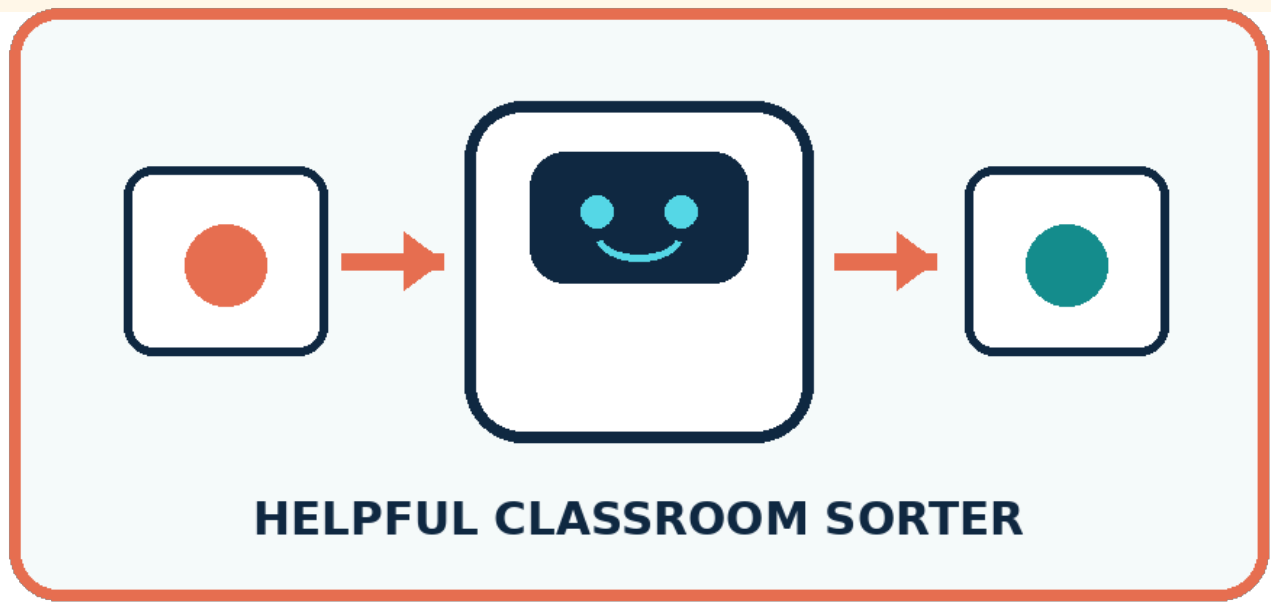
RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

11. Connect your project

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one evidence-based decision at a time.



THIS WEEK'S CHECKPOINT Complete the sorter's human-control map: permissions, review triggers, override, correction, and stop conditions.

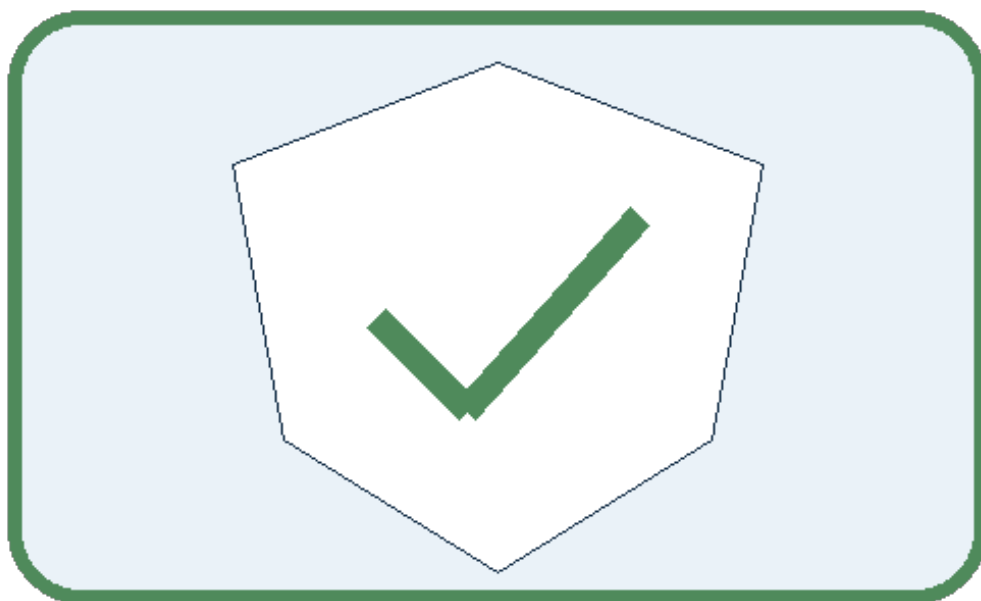
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the process?

What normal and edge cases will you test?

When should it say not sure or ask a person?



RESPONSIBLE DESIGN A human in the room is not meaningful oversight if the person cannot understand or change the outcome.

My project decision:

Evidence for your project

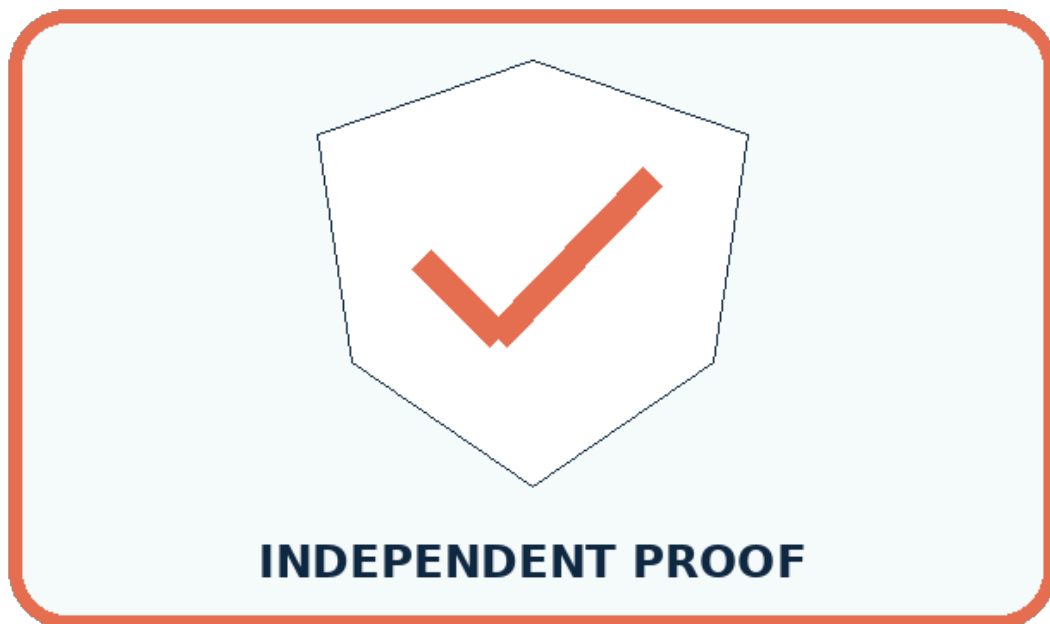
Record sorter actions allowed automatically and actions requiring the classroom owner's explicit review.

12. Show what you know

MASTERY

Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



REQUIRED PROOF Add adult-review gates to a classroom helper and explain the permission, evidence, and consequence at each gate.

- ☐ Name or draw the fresh case.
- ☐ Trace input, process, and output.
- ☐ State the conclusion and strongest evidence.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent evidence:

SELF-CHECK Mark where data enters, where the system predicts, where a person reviews, and who may act. Give reviewers the evidence and power to correct or stop the process. High-consequence cases should not pass automatically.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

NEXT BRIDGE Carry today's evidence habit into: Find a Real Problem.

13. Study a complete case

Week 28 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

An assistant may suggest object locations; a teacher must approve moving valuable equipment.

The question

What may happen without approval?

Worked explanation

It may suggest a location within its role. Moving equipment requires actual teacher approval.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

A suggestion must stop at the approval gate

Evidence or stage · Value or result

Evidence or stage: Assistant

Value or result: Suggest equipment location

Evidence or stage: Teacher

Value or result: Review and decide

Evidence or stage: Approval

Value or result: Explicit permission to move

Evidence or stage: Action

Value or result: Move only after approval

An assistant may suggest object locations; a teacher must approve moving valuable equipment.

What may happen without approval?

It may suggest a location within its role. Moving equipment requires actual teacher approval.

14. Try a new case independently

Independent case: Transfer the permission check to a new case

In a fictional library, the librarian controls the internal shelf list. The librarian tells assistant Jo: "Replace the internal list with checked version L2 on Friday, once. Do not change the public catalogue." It is Friday, L2 remains checked, and no replacement has occurred. A tool offers both destinations. Assess these proposals separately: Jo replaces the internal list with L2; Jo updates the public catalogue with L2; visitor Sam replaces the internal list with L2; Jo replaces the internal list with a changed L3. Identify the approval detail that settles each case.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 28 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A sorter marks a child's notebook throw away. Who should decide whether it is discarded? Name a boundary the system should enforce.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Complete the sorter's human-control map: permissions, review triggers, override, correction, and stop conditions.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

A classroom assistant may draft a reminder but cannot send messages. It displays a completed draft and waits. A student writes 'looks good' without a destination. Has a clear send authorisation been given? Specify what is needed.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: A fictional teacher approves one checked classroom poster for internal display.

By the end, I can · How I will show it

By the end, I can: Distinguish a system's capability, the correctness of its output and permission to take an action

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Match an action to the person, version, audience and limits covered by an approval

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Explain when to proceed within an existing permission and when a changed action needs review

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Distinguish a system's capability, the correctness of its output and permission to take an action

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional teacher approves one checked classroom poster for internal display.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Public-web publication exceeds the approved destination.

1. Allowed action
2. Changed audience
3. Read authority
4. Changed version

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Changed audience Public-web publication exceeds the approved destination. A general intention to help the class does not grant every action that might help. Check the version, destination and time explicitly. An edit may improve accuracy while still changing the approved artifact. The assistant should hold the changed poster for review rather than treating its own confidence as permission.

Practice 2: explain the deciding evidence

Changed-case practice: P2 has already been published once under this approval. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. A second publication is not covered.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: A second publication is not covered. The instruction allowed a single use, not repeated actions.

Repair a missing building block

Review Verify Important Outputs (ai-1/27). Goal: Separate an answer into claims and connect each claim to relevant evidence. Short practice: Correct the draft and identify which part needs another source rather than a calculation.. Return to this week and explain what you changed.

Review Generate, Retrieve, or Calculate? (ai-1/25). Goal: Distinguish retrieving a record, calculating from inputs and generating wording. Short practice: Label each arrow with its operation. Can the system add snacks because invitations often mention them?. Return to this week and explain what you changed.

Review Missing and Unbalanced Examples (ai-1/22). Goal: Name the target collection, selection list and cases actually assessed. Short practice: Would testing only dim plain cards close every current gap? Point to the target groups.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.